

# Dhrumil Rana

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## EXPERIENCE

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### Founding AI Systems Engineer | *NovaChatAI*

March 2026 to Present

- Own architecture and delivery across **8+ production product surfaces** spanning conversational AI, messaging, scripts, PPV monetization, media, subscribers, campaigns, analytics, and authentication using **Python, FastAPI, PostgreSQL, Supabase, Next.js, and TypeScript**.
- Resolved **100+ production defects and functionality chokepoints** across frontend, backend, database, AI execution, authentication, background workers, external APIs, and monetization while continuously shipping new product functionality.
- Eliminated duplicate and conflicting automation execution through **persistent fan state, idempotency guards, database enforced uniqueness, scheduler lifecycle controls, and routing fallbacks** across asynchronous AI workflows.
- Engineered **stateful AI conversation and monetization flows** across relationship building, scripts, PPV, follow ups, and re engagement by combining conversation state, purchase context, routing logic, memory, and execution safeguards so model decisions translate into controlled product actions.
- Architected relational systems for **mass messaging, campaign automation, script execution, and monetization state**, modeling recipients, delivery attempts, events, scheduling, recovery, and execution history with foreign keys, partial indexes, RLS, and safe migrations.
- Built **recovery controls for scheduled automation** using claim based execution, pending state indexes, attempt history, internal service authorization, and regression tests so partial failures resume safely.
- Diagnosed a production AI response outage with **9,000+ queued jobs** by tracing cron concurrency, worker saturation, queue state, and database records; restored processing and verified successful backlog drainage.
- Reworked authorization around **session derived identity, ownership validation, private PostgreSQL schemas, row level security, and scoped roles** while owning failures end to end across React, Next.js, FastAPI, AI orchestration, PostgreSQL, workers, OAuth, and external integrations.

### Software Developer Intern | *Odoo*

May 2024 to September 2024

- Developed **Python backend modules, REST APIs, and PostgreSQL business logic** for point of sale, inventory, and enterprise workflows while resolving **50+ production defects** across application logic, databases, and integrations.
- Analyzed database access patterns and optimized **PostgreSQL queries and backend execution paths**, improving transaction performance by **15%** across production business workflows.

### Research Analyst | *Silent Info Tech Pvt. Ltd.*

January 2024 to April 2024

- Built an end to end **Python ETL and analytics pipeline** processing **100K+ records across 20+ features**, transforming raw marketing data into validated datasets for downstream analysis and reporting.
- Integrated **OpenAI language models** into content processing and transformation workflows, reducing measured manual processing effort by **80%**.

## PROJECTS

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### Aegis Federate | *Privacy Preserving Federated Intelligence*

- Architected a **cross silo federated learning system** using **Flower, gRPC, Docker, PyTorch, 1D CNNs, and MLPs**, allowing heterogeneous models to learn collaboratively without centralizing raw data.
- Implemented **differential privacy with DP SGD and Opacus**, combining gradient clipping and controlled noise to evaluate privacy and predictive utility tradeoffs.

### Agentic Cleanse | *Autonomous Data Quality Engine*

- Built a **stateful multi agent data engineering system** with **LangGraph, Python, and large language models** for profiling, anomaly detection, schema reasoning, transformation, validation, and correction.
- Converted probabilistic model reasoning into controlled ETL operations with deterministic validation, reducing manual data processing effort by **80%**.

### Twitter Scale Link Prediction Engine | *Graph Machine Learning*

- Engineered a graph learning pipeline across **1.2M nodes**, generating **1M+ DeepWalk sequences** and a **1,028 dimensional representation** combining embeddings with structural graph signals.
- Trained a Random Forest link prediction model achieving **0.887 ROC AUC**, outperforming heuristic only approaches through topology, neighborhood statistics, and representation learning.

## EDUCATION

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### Master of Science in Data Science | *University of Texas at Arlington*

January 2025 to July 2026

### Bachelor of Engineering in Information Technology | *Government Engineering College Modasa*

November 2020 to May 2024

## TECHNICAL SKILLS

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- **AI and Machine Learning:** Generative AI, RAG, LangGraph, LangChain, PyTorch, TensorFlow, Scikit learn, graph ML, federated learning, Flower, Opacus
- **Backend and Data:** Python, FastAPI, SQL, PostgreSQL, Supabase, REST APIs, schema design, indexing, migrations, workers, schedulers, state machines, idempotency, gRPC
- **Product and Analytics:** TypeScript, React, Next.js, Docker, Git, OAuth, RLS, API integrations, Pandas, NumPy, ETL, Tableau, Power BI